Knowledge Unlearning using Sparse AutoEncoders

TLDR; We want LLM's to forget Harry Potter (& wizarding universe)

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Defining the Problem

IP (Intellectual Property) Memorization in LLMs

Large Language Models (LLMs) often inadvertently memorize copyrighted intellectual property (IP) during pre-training, leading to ethical and legal risks.

Traditional unlearning methods rely on weight modification, which lacks transparency and risk "catastrophic forgetting" of general knowledge.



Background & Existing Methods

Approximate Unlearning

Eldan & Russinovich (2023) proposed fine-tuning to forget content by replacing idiosyncratic terms with generic alternatives.

1. Detect Harry Potter related tokens
2. Compare logits with baseline model
3. Replace specific phrases with generic ones
4. Generate alternative next token predictions
5. Finetune model using new labels



Limitations

- Unlearning may affect global behavior
- May forget relevant non-target information
- Hard to isolate exact knowledge region

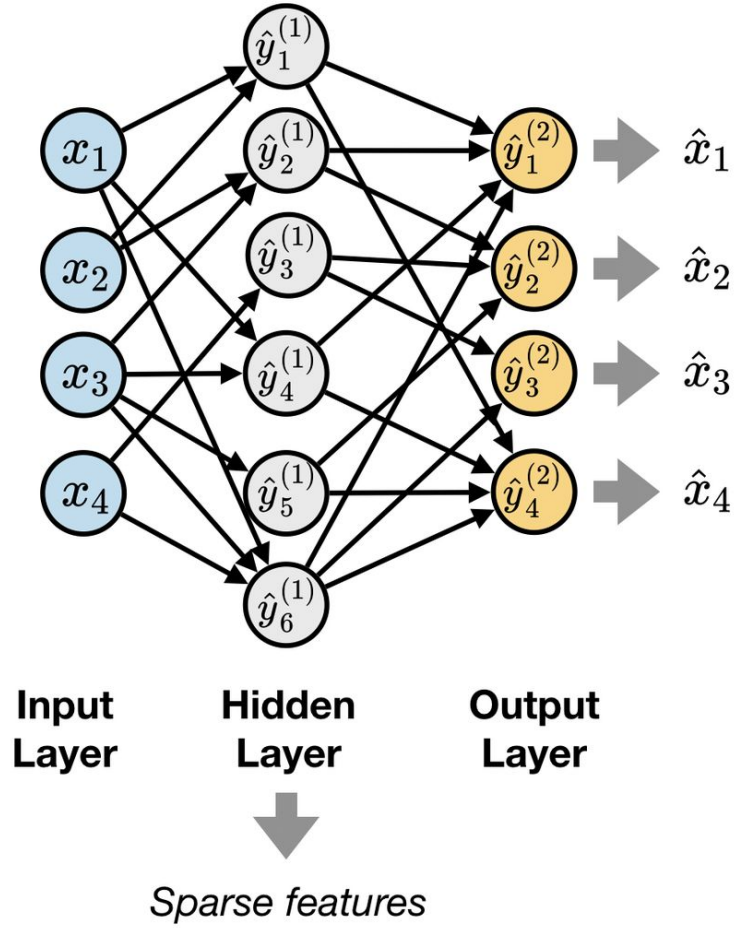
Sparse Autoencoders (SAE)

What is an SAE?
A Sparse Autoencoder is a model that learns to represent activations as a **small set of meaningful features**, instead of dense neuron mixtures.

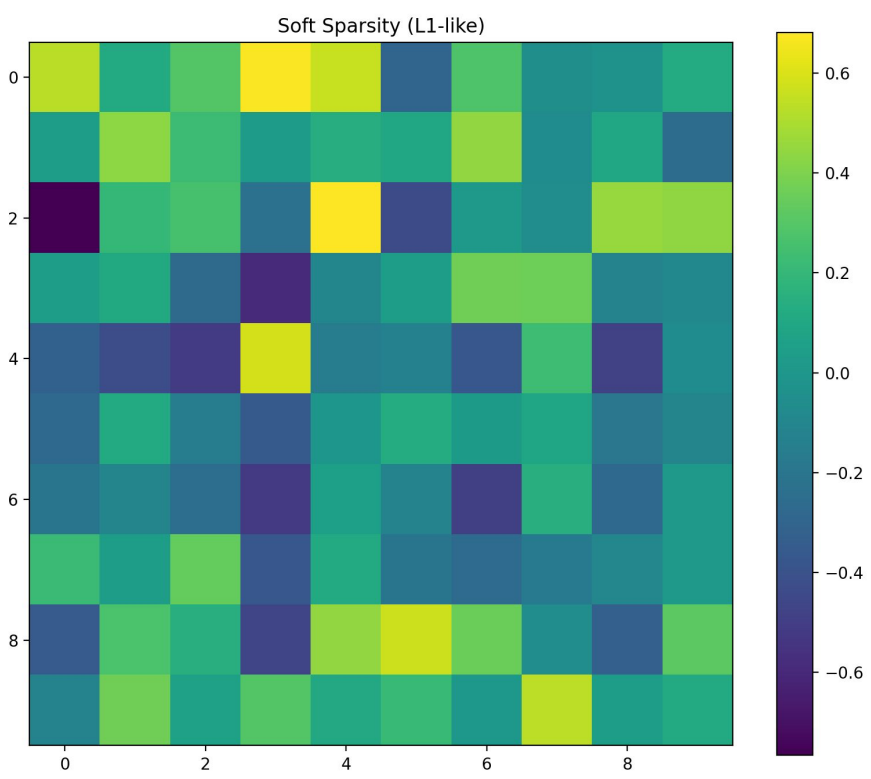
How it works (pipeline)
Input activations x -> **encoder** -> sparse features z (few active) -> **decoder** reconstructs x.

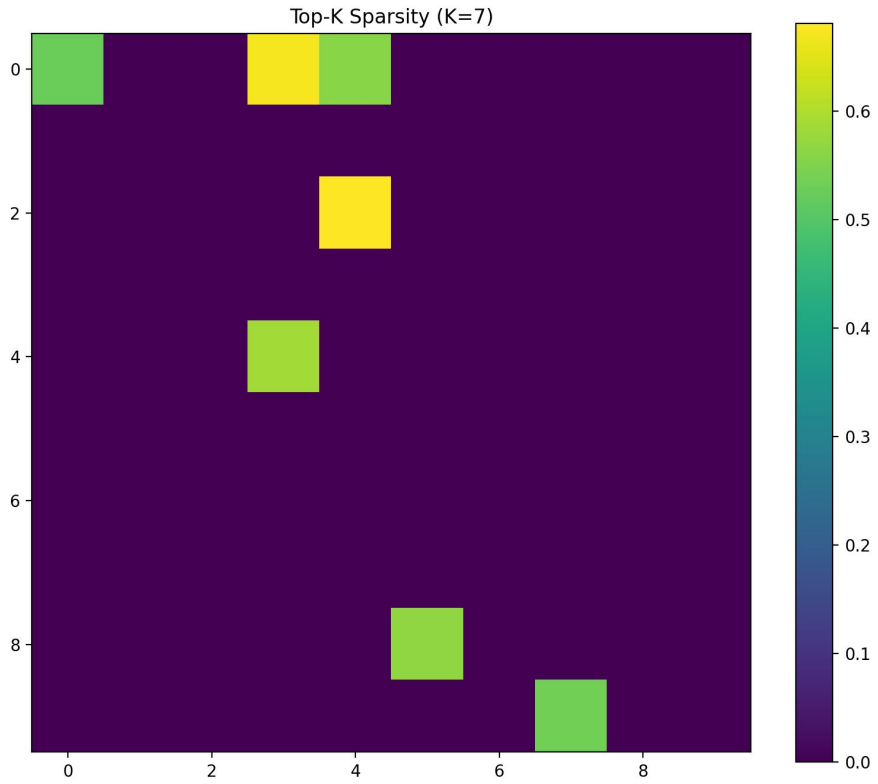
Key idea (sparsity)
A sparsity penalty forces only a few features to fire, so each feature captures a **distinct concept** (e.g., a “Harry Potter” feature).

Why useful here
Because features are clean and separable, you can **identify and selectively suppress specific concepts** (like HP) without damaging the rest of the model.



Enforcing Sparsity via Top-K





Problem with Normal Sparsity

- Encourages sparsity but does not enforce it
- Many neurons remain slightly active
- Leads to overlapping and hard to interpret features

Why Top K Sparsity

- Keeps only K activations and sets others to zero
- Ensures true sparsity
- Produces clear and interpretable features

Why SAE is a Better Approach



Interpretability

Ablated features are human-inspectable. We verify exactly what is removed.



No Weight Edits

Intervention via inference-time hooks. Base model remains intact and reversible.



High Precision

Minimal damage to adjacent capabilities (Magic vs. Harry Potter concepts).

Experimental Setup

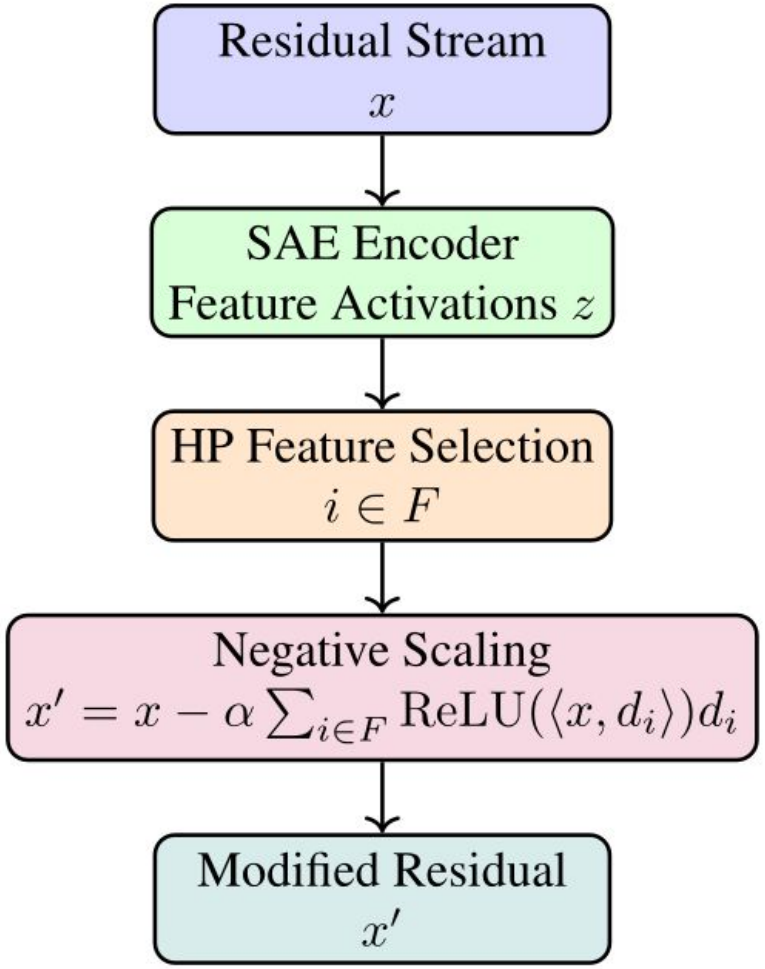
Component	Specification
Base Model	GPT-2 Medium (355M Parameters)
Target Layer	Layer 12 Residual Stream (d=1024)
Target Corpus	Harry Potter Book Series (1.3M Tokens)
Neutral Corpus	WikiText-2 + TinyStories (2.8M Tokens)
Evaluation	LLM-as-Judge, Log-Prob Shifts, PPL

Implementation Details

- **Libraries:** PyTorch for SAE training and **TransformerLens** for mechanistic interventions.
- **Layer Selection:** Layer 12 selected as semantic world knowledge is more linearly separable in deeper layers.
- **Regularization:** Unit-Norm Decoder constraint and Gradient Adjustment for stable dictionary learning.
- **Post-Ablation:** Top-p sampling (p=0.9) and Frequency Penalty (1.0) applied during autoregressive decoding.

Visual Pipeline

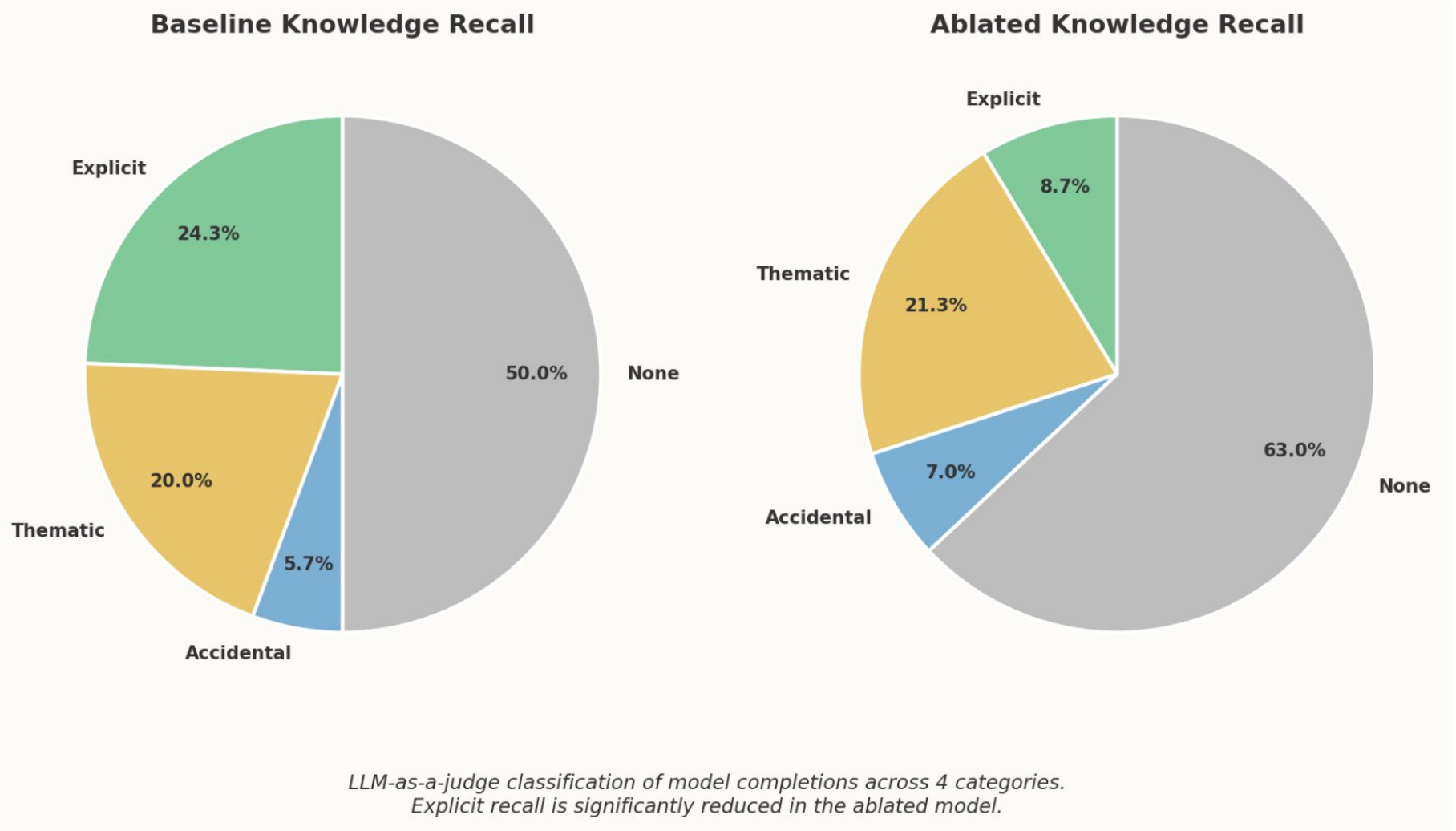
- **Hook layer 12 activations**: Intercept the residual stream at layer 12 during the forward pass.
- **Use SAE to find "Harry Potter" features**: Encode activations into sparse features and identify those linked to Harry Potter concepts.
- **Select those features**: Choose the subset FFF corresponding to Harry Potter-related neurons/features.
- **Suppress them**: Subtract (scale down) their contribution from the activations using the negative scaling step.
- **Continue forward pass**: The modified activations X'' flow through the model → it **forgets or avoids Harry Potter knowledge** in outputs.



Quantitative Metrics

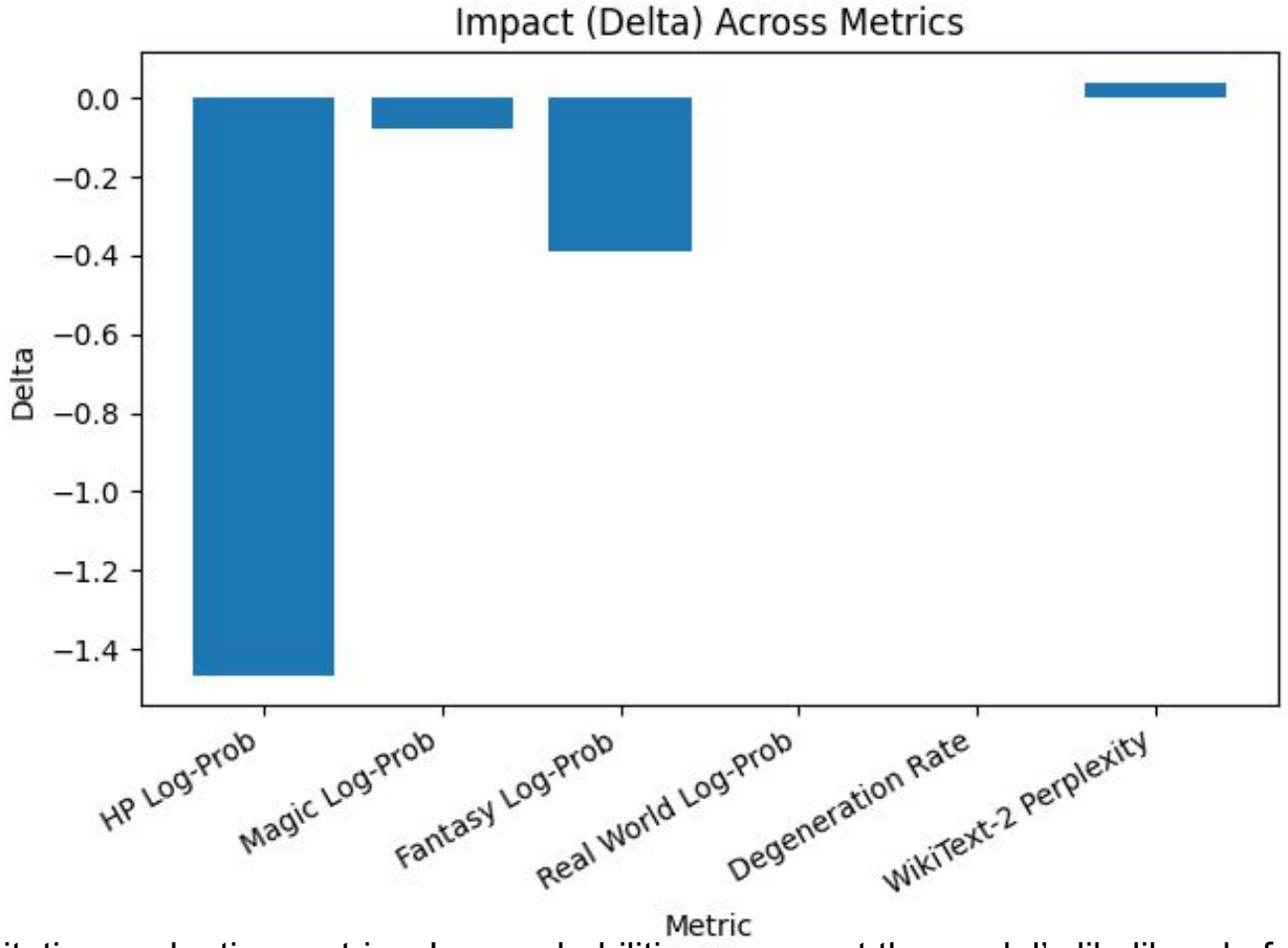
Metric	Baseline (Top-P)	Ablated (Scale -3.0)	Impact (Δ)
Target Knowledge			
HP Log-Probability	-3.6567	-5.1248	$\mathbf{-1.4681}$ **(Massive Drop)**
Surgical Selectivity			
Magic Log-Probability	-3.0521	-3.1354	Preserved (-0.08)
Fantasy Log-Probability	-6.4321	-6.8215	Minor Collateral (-0.39)
Real World Log-Prob	-3.9647	-3.9692	Unchanged (≈ 0.00)
Capability Stability			
Degeneration Rate	0.0%	0.0%	Perfectly Stable
WikiText-2 Perplexity	43.84	43.88	0.1% Degradation

Knowledge Suppression Results



LLM-as-a-judge classification of model completions across 4 categories.
Explicit recall is significantly reduced in the ablated model.

Knowledge Suppression Results



Quantitative evaluation metrics. Log-probabilities represent the model's likelihood of predicting domain-correct tokens. A more negative value in the HP domain indicates successful erasure.

Conclusion

Interpretable, Surgical, and Reversible Unlearning

SAEs provide a powerful framework for surgical IP unlearning. By targeting specific features, we achieve knowledge erasure with negligible impact on model stability.

